

Task 23 – SQL Window Functions Intro

Data Analytics Track | Day 23

1. Objective

The objective is to learn analytical SQL patterns using ROW_NUMBER, RANK, DENSE_RANK and LAG for common business questions such as ranking sales, identifying top transactions and comparing current values with previous values.

2. Dataset

A synthetic sales dataset containing 36 transactions, 6 customers, 2 categories and 5 products was created so that the window functions can be demonstrated with clear business-oriented examples. The dataset includes sale date, customer, category, product, quantity, unit price and sales amount.

3. Window Functions Used

Function	Purpose	Business Example
ROW_NUMBER()	Assigns a unique sequential number	Order each customer's sales
RANK()	Ranks values; ties create gaps	Rank highest sales
DENSE_RANK()	Ranks values; ties do not create gaps	Rank products within a category
LAG()	Reads a previous row's value	Compare a sale with the previous sale

4. Twelve Queries Completed

Query	Window Function	Business Purpose
Q01	ROW_NUMBER	Overall sales ordering
Q02	ROW_NUMBER + PARTITION	Order sales within each customer
Q03	RANK	Overall sales ranking
Q04	DENSE_RANK	Overall dense ranking
Q05	RANK + PARTITION	Category-wise sales ranking
Q06	DENSE_RANK + PARTITION	Customer-wise dense ranking
Q07	LAG + PARTITION	Previous customer sale
Q08	LAG	Change from previous customer sale
Q09	LAG	Previous overall transaction
Q10	ROW_NUMBER + CTE	Top 2 sales per customer
Q11	DENSE_RANK + CTE	Top-ranked sales per category
Q12	LAG + aggregation	Month-over-month sales change

5. Key Findings from the Outputs

Monthly sales totals were calculated with a GROUP BY and then compared using LAG. The resulting output is included in Q12 and in the Excel workbook.

Top 5 transactions by sales amount:

Sale ID	Customer	Product	Sales Amount
13	Alice	<bound method Series.prod of sale_id sale_date 2026-03-04 00:00:00 customer Alice category Electronics product Laptop quantity 3 unit_price 800 sales_amount 2400 Name: 12, dtype: object>	2,400
25	Alice	<bound method Series.prod of sale_id sale_date 2026-05-03 00:00:00 customer Alice category Electronics product Monitor quantity 8 unit_price 220 sales_amount 1760 Name: 24, dtype: object>	1,760
1	Alice	<bound method Series.prod of sale_id sale_date 2026-01-05 00:00:00 customer Alice category Electronics product Laptop quantity 2 unit_price 800 sales_amount 1600 Name: 0, dtype: object>	1,600
10	Diana	<bound method Series.prod of sale_id sale_date 2026-02-16 00:00:00 customer Diana category Electronics product Laptop quantity 2 unit_price 800 sales_amount 1600 Name: 9, dtype: object>	1,600
20	Bob	<bound method Series.prod of sale_id sale_date 2026-04-07 00:00:00 customer Bob category Electronics product Laptop quantity 2 unit_price 800 sales_amount 1600 Name: 19, dtype: object>	1,600

6. Interview Notes

RANK vs DENSE_RANK: Both assign the same rank to tied values. RANK leaves gaps after a tie, while DENSE_RANK continues without gaps. For values 100, 100, 90, RANK produces 1, 1, 3 and DENSE_RANK produces 1, 1, 2.

When to use LAG: LAG is useful when the current row must be compared with a previous row, for example previous-month sales, previous purchase amount or transaction-to-transaction change.

7. Deliverables Included

Item	Included
12 SQL queries	Yes
Query outputs	12 CSV files + Excel workbook
Source dataset	CSV + SQLite database
Notes	NOTES.md
PDF report	Yes